MT CLASSES
S.K. Nagar, Motihari • CBSE Maths & Science, Class 9–10

AC & DC Current and Domestic Electric Circuits

Class 10 Science | Chapter: Magnetic Effects of Electric Current | Chapter Notes

1. Electric Current — Quick Recap

Electric current is the rate of flow of electric charge through a conductor: I = Q / t. Its SI unit is the **ampere (A)**. Depending on how the current flows with time, it is of two types: **Direct Current (DC)** and **Alternating Current (AC)**.

2. Direct Current (DC)

A current that flows in **one direction only** and whose magnitude usually stays constant is called **direct current (DC)**.

Sources of DC

- Electric cells and batteries (dry cell, torch cell, car battery)
- Solar cells
- DC generator (dynamo)

Uses of DC

- Charging mobile phones, laptops and inverter batteries
- Torches, remote controls, wall clocks
- Electroplating and electrolysis (need fixed direction of current)

3. Alternating Current (AC)

A current that **reverses its direction periodically** (after equal intervals of time) is called **alternating current (AC)**. Its magnitude also changes continuously, rising from zero to maximum and falling back.

Sources of AC

- AC generator (works on electromagnetic induction)
- Power stations — thermal, hydro, nuclear — all produce AC

> **Key Facts about AC in India**
>
> **Frequency = 50 Hz** — AC completes 50 cycles every second.
>
> **Time for one cycle = 1/50 s = 0.02 s.**
>
> In one cycle the direction reverses twice, so the current **changes direction every 1/100 s** (i.e. 100 times per second).
>
> **Household supply voltage = 220 V.** (USA uses 110–120 V at 60 Hz.)

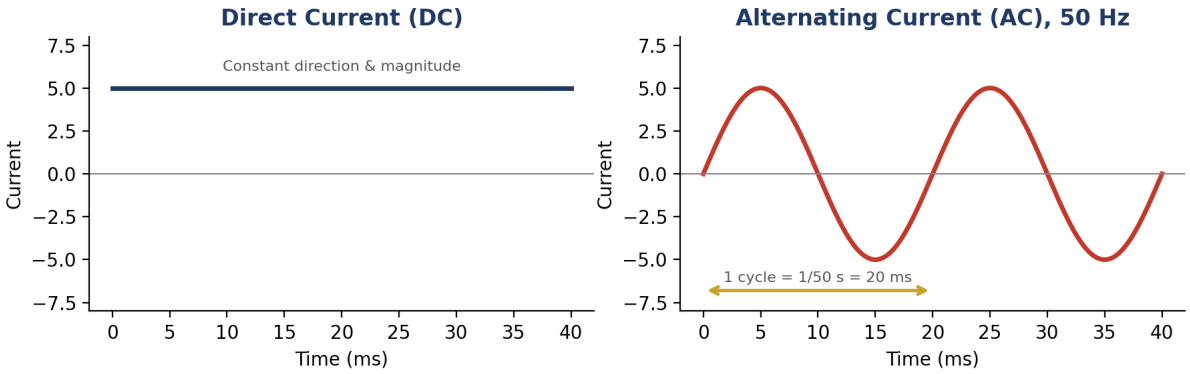


Fig. 1 — Graph of current vs time for DC (left) and AC (right)

4. Difference Between AC and DC

Basis	Alternating Current (AC)	Direct Current (DC)
Direction	Reverses periodically	Always in one direction
Magnitude	Changes continuously with time	Generally constant
Frequency	50 Hz in India	Zero
Source	AC generator, power stations	Cells, batteries, DC generator
Transmission	Can be sent over long distances with less energy loss	Large energy loss over long distances
Voltage change	Easily stepped up/down by a transformer	Cannot be changed by a transformer
Storage	Cannot be stored in a battery	Can be stored in a battery
Graph	Wave (sine curve)	Straight line parallel to time axis

> **Why is AC preferred for supply to homes?**
>
> The main advantage of AC over DC is that **AC can be transmitted over long distances without much loss of energy**. Its voltage can be raised with a step-up transformer at the power station (so current and heating losses in the wires are low) and lowered again with a step-down transformer near our homes.

5. Domestic (Household) Electric Circuits

Electricity reaches our homes from the pole or underground cable through the **mains supply** at **220 V AC, 50 Hz**. A household circuit uses **three wires**:

Wire	Old colour	New (IEC) colour	Potential / Role
Live (positive) wire	Red	Brown	At 220 V; carries current into the appliance
Neutral (negative) wire	Black	Light blue	At 0 V; completes the circuit (return path)
Earth wire	Green	Green-yellow	Connected to a metal plate buried deep in the earth; safety wire

The potential difference between the live and neutral wire in India is **220 V**.

Path of current in the house

1. **Main fuse** — installed by the electricity board at the entry point; protects the whole house.
2. **Electricity meter** — records the electrical energy consumed in kilowatt-hour (kWh or 'units').

3. **Main switch and distribution box** — the supply is divided into separate circuits for different rooms, each with its own fuse or MCB.

Two separate circuits are used

Circuit	Current rating	Appliances
Low-power circuit	5 A	Bulbs, tube lights, fans
High-power circuit	15 A	Geyser, room heater, electric iron, AC, refrigerator, washing machine

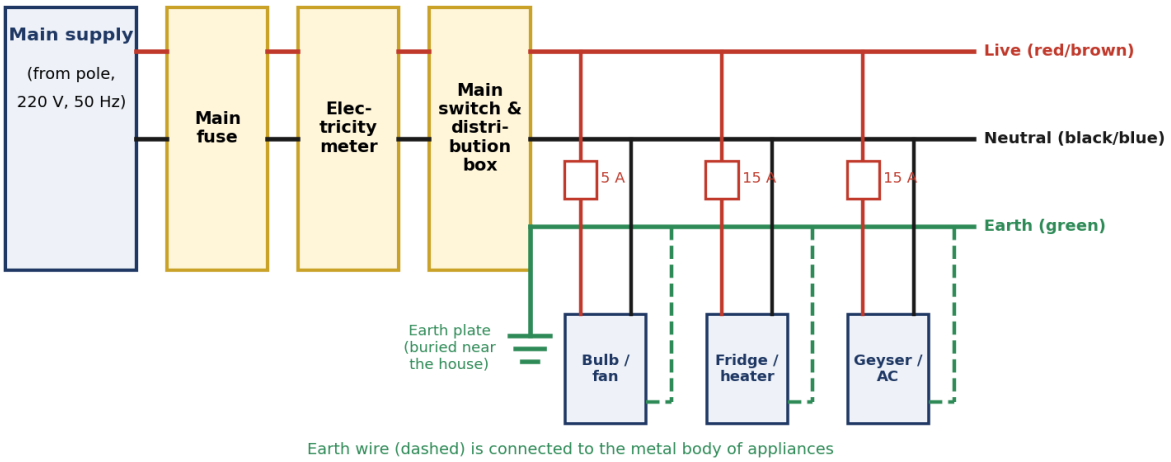


Fig. 2 — Schematic diagram of a domestic electric circuit

Why are appliances connected in parallel?

- Each appliance gets the **same voltage (220 V)**.
- Each appliance can be **switched ON/OFF independently** using its own switch.
- If one appliance fails or is switched off, **others keep working**.
- Total resistance is less, so each appliance draws the **current it needs**.

6. Earthing (Earth Wire)

Appliances with a **metallic body** (electric iron, toaster, refrigerator, geyser, washing machine) are connected to the **earth wire**. It is joined to a **metal plate buried deep in the ground** near the house.

> **Purpose of Earthing**
>
> If the live wire accidentally touches the metal body due to faulty insulation (**leakage of current**), the earth wire provides a **low-resistance path** for the current to flow into the earth.
>
> This keeps the potential of the metal body **equal to that of the earth (zero)**, so the user **does not get a severe electric shock**.

7. Faults in Household Circuits

(a) Short Circuit

A **short circuit** occurs when the **live wire and neutral wire come in direct contact**, usually because the insulation is damaged or the appliance is faulty. The resistance of the circuit becomes almost zero, so a **very large current** flows (I = V/R). This produces excessive heat and can cause **sparks and fire**.

(b) Overloading

Overloading occurs when **too many appliances are connected to a single socket**, or the supply voltage rises accidentally. The total current drawn exceeds the safe limit of the wires, which overheat and can catch fire.

> **Remember**
> Short circuit → live and neutral touch → resistance ≈ 0 → current becomes very high.
> Overloading → too many appliances in one socket → total current exceeds safe limit.
> Both can cause fire; the **fuse or MCB** protects against both.

8. Electric Fuse and MCB

An **electric fuse** is a safety device that protects circuits and appliances from damage due to short-circuiting or overloading.

- **Principle:** Heating effect of electric current (Joule's heating). When excess current flows, the fuse wire gets heated, **melts and breaks the circuit**.
- **Material:** An alloy of **tin and lead** (or copper/aluminium in modern fuses) with a **low melting point** and suitable resistance.
- **Connection:** Always connected **in series with the live wire**, so the appliance is cut off from the high-potential wire when it blows.
- **Ratings:** Fuses are rated 1 A, 2 A, 3 A, 5 A, 10 A, 15 A etc. A fuse of **slightly higher rating than the normal current** of the appliance is used.
- **MCB (Miniature Circuit Breaker):** An automatic switch that trips (turns off) when the current exceeds a safe value. Unlike a fuse, it **need not be replaced** — it can simply be switched on again after the fault is repaired.

9. Solved Numericals

Q. An electric oven of 2 kW power rating is operated in a domestic circuit (220 V) that has a current rating of 5 A. What result do you expect?

Ans. P = V × I ⇒ I = P / V = 2000 / 220 = **9.09 A**. The current drawn (9.09 A) is more than the rating of the circuit (5 A), so the **fuse will melt / MCB will trip** and the circuit will break. The oven should be used on the 15 A circuit.

Q. What fuse should be used for an electric iron of 1 kW operating at 220 V?

Ans. I = P / V = 1000 / 220 = **4.54 A**. A fuse of slightly higher rating is needed, so a **5 A fuse** is suitable.

Q. How many times does the direction of AC in India change in one second?

Ans. Frequency = 50 Hz, and the direction changes twice in every cycle. So it changes **2 × 50 = 100 times per second**, i.e. once every 1/100 s.

10. Electrical Safety Measures

- Use wires of good quality with proper insulation and correct current rating.
- Always use a fuse or MCB of proper rating in the live wire.

- Connect metallic appliances to the earth wire (three-pin plug).
- Do not connect too many appliances to one socket; avoid multi-plug overloading.
- Switch off the main supply before any repair; never touch switches with wet hands.

11. Important Questions (Board Pattern)

Q. What is the difference between direct current and alternating current? Name one source of each. (2 marks)

Ans. DC flows in one direction only; AC reverses its direction periodically. Source of DC: electric cell/battery. Source of AC: AC generator/power station.

Q. Name the wire that is at zero potential in a household circuit.

Ans. The **neutral wire**.

Q. Why is a fuse connected in series in the live wire?

Ans. So that when excess current flows, the fuse melts and the appliance gets disconnected from the high-potential live wire, making it safe.

Q. What is the function of an earth wire? Why is it necessary to earth metallic appliances? (3 marks)

Ans. The earth wire provides a low-resistance conducting path for current. If current leaks to the metal body of an appliance, it flows to the earth, keeping the body at earth potential, so the user does not get a severe shock.

Q. What precaution should be taken to avoid overloading of domestic circuits?

Ans. Do not connect too many appliances to a single socket, and use a fuse/MCB of the correct rating in the circuit.

Q. Why is AC preferred over DC for long-distance transmission?

Ans. AC can be transmitted over long distances with little loss of energy because its voltage can be stepped up and down easily with transformers.

Q. Why are copper or aluminium wires used for electrical transmission and not fuse wire alloy?

Ans. Copper and aluminium have very low resistivity and high melting points, so they carry current with little heating. Fuse wire has a low melting point and is designed to melt.

Quick Revision

One-line points

DC → one direction; AC → reverses direction periodically.
India: 220 V, 50 Hz; AC changes direction every 1/100 s.
Main advantage of AC: long-distance transmission with less energy loss.
Live (red/brown) 220 V; Neutral (black/blue) 0 V; Earth (green) — safety.
5 A circuit → bulbs & fans; 15 A circuit → high-power appliances.
Household appliances are connected in parallel.
Short circuit: live touches neutral. Overloading: too many appliances in one socket.
Fuse: works on heating effect; low melting point; in series with live wire.